Probing Internal Signals of Topic, Difficulty, and Prediction of Success in a 7B Math LLM

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1. Goal & Motivation

We will evaluate a single open-weight 7B math-specialized LLM on competition-style problems and train linear probes on frozen activations to test three questions:
(O2) *Topic probe:* is the problem topic (algebra/number theory/combinatorics) linearly decodable after reading the question?
(O3) *Success probe:* can a probe predict eventual correctness from the model's state at 0% (question-only) and 50% (mid-solution replay)? (we may extend this to 25% and 75% progress points as well if time permits)
(O4) *Difficulty probe:* do activations encode human difficulty (MATH levels 1–5), and how does that align with actual accuracy?
These interpretability checks connect to evidence that simple linear directions can track truthfulness [1] and in-advance correctness [2], and extend them to math reasoning where models have shown 'cliffs' of performance within certain problem types.

Importance: If an LLM internally "knows" topic, senses hardness, or foresees failure mid-solution, we can design routing/guardrails (e.g., self-consistency, fallback to tools, human-in-the-loop) to reduce wrong-but-confident answers, most crucially within areas where failure and uncertainty is high.

2. Related Work

Linear probes & emergent linear structure. Linear classifier probes [5] reveal what information is linearly decodable at each layer. Recent work shows linear structure for truth vs. falsehood in LLM residual streams and provides causal evidence via interventions [1]. Closest to our success probe, question-only linear probes can predict whether the forthcoming answer will be correct, with signals peaking at mid layers, but **generalization is weaker on math, motivating our domain-specific study** [2].

Math benchmarks & inference strategies. The MATH benchmark by Hendrycks provides 12.5k competition problems with topic tags and difficulty levels (1–5), plus solutions [3]. Inference-time self-consistency (sample multiple chains, then vote) reliably boosts math accuracy and motivates our multi-sample evaluation [4].

Models & tooling. We will select an open 7B model tuned for math (e.g., DeepSeekMath-7B, Mathstral-7B) that cleanly exposes hidden states for probing [6–7], and use TransformerLens/HF hooks to capture activations [8]. Note: initial testing with determine our model of choice here.

3. Method

(Objective 1) Performance matrix by topic × difficulty (How well does the model perform at different topics of different difficulties?)

- **Dataset:** Stratified MATH subset (≈400–600 problems) balancing topics and difficulty (1–5). We will exclude geometry items, given issues with describing such problems to text only models.
- **Decoding:** Given resource constraints we propose using 5 samples/problem, but may adjust this if variance is high. Our prompt enforces a standard final answer format (e.g., \boxed {answer}) so an additional evaluation model or scheme is not necessary.
- **Outputs:** Accuracy vs. difficulty and topic, and identification of steep drops ("cliffs").

(Objective 2) Topic probe (Given the question and before any output, do the hidden states of the model tell us it knows what type of problem is being asked?)

- **Inputs:** Hidden states after encoding the question (no generated tokens). ● **Features:** Start with final token residual per layer (or mean-pooled question tokens). ● **Layer sweep:** Evaluate every 4th layer first, then zoom into the best region.

(Objective 3) Success probe [0%,50%] (How well can the model before answering as well as halfway through answering, predict its correctness?)

- **Labeling:** Generate once to obtain an answer and ground-truth correct/incorrect.
- **Checkpoints (replayed back after initial generation):**
    - **0%:** question only.
    - **50%:** question + first half of the *already generated* tokens.
- **Probe:** Logistic regression (balanced classes; split by problem). This tests whether

internal states anticipate success and whether the signal strengthens mid-solution [2].

(Objective 4) Difficulty probe (question-only, may exclude if time and resources are insufficient)

- Predict human difficulty (regression or 5-way classification) from question-only states; correlate with actual accuracy. Interesting cases: predicted "easy" yet wrong (overconfidence) vs. "hard" yet right (good, may provide predictive power that model is nearing its capability limit)

Considerations:

- **VRAM limits:** We're aiming to run on a 16 GB card in FP16/BF16 (but may quantize if ending up with insufficient space). In addition, to reduce VRAM needs we will restrict context (≤1.5k tokens).
- **Data hygiene:** Train/val/test split by problem, we will balance classes within our selected subset of the MATH dataset, and we will store seeds and prompts for reproducibility

4. Proposed Framework

Our framework investigates how a 7B math-specialized LLM internally represents problem topic, difficulty, and likelihood of success through simple linear probes on frozen activations. Using a stratified subset of the MATH dataset, we test whether interpretable, linearly decodable signals emerge across layers of the model.

We focus on three complementary probes:

1. **Topic Probe:** Tests if the model distinguishes between algebra, number theory, and combinatorics after reading the question.
2. **Success Probe:** Evaluates whether hidden states encode the model's own likelihood of answering correctly at 0% and 50% checkpoints.
3. **Difficulty Probe:** Measures how internal representations align with human difficulty ratings and observed accuracy.

This framework extends prior probing work beyond linguistic or factual reasoning into structured mathematical problem solving, where sequential logic plays a key role. By using lightweight logistic and linear regression probes without fine-tuning, our approach remains simple, efficient, and interpretable. The resulting analyses connect performance outcomes to internal representations, offering insight into whether the model "knows" what kind of problem it is solving and how confident it should be.

5. Current Experiments and Results

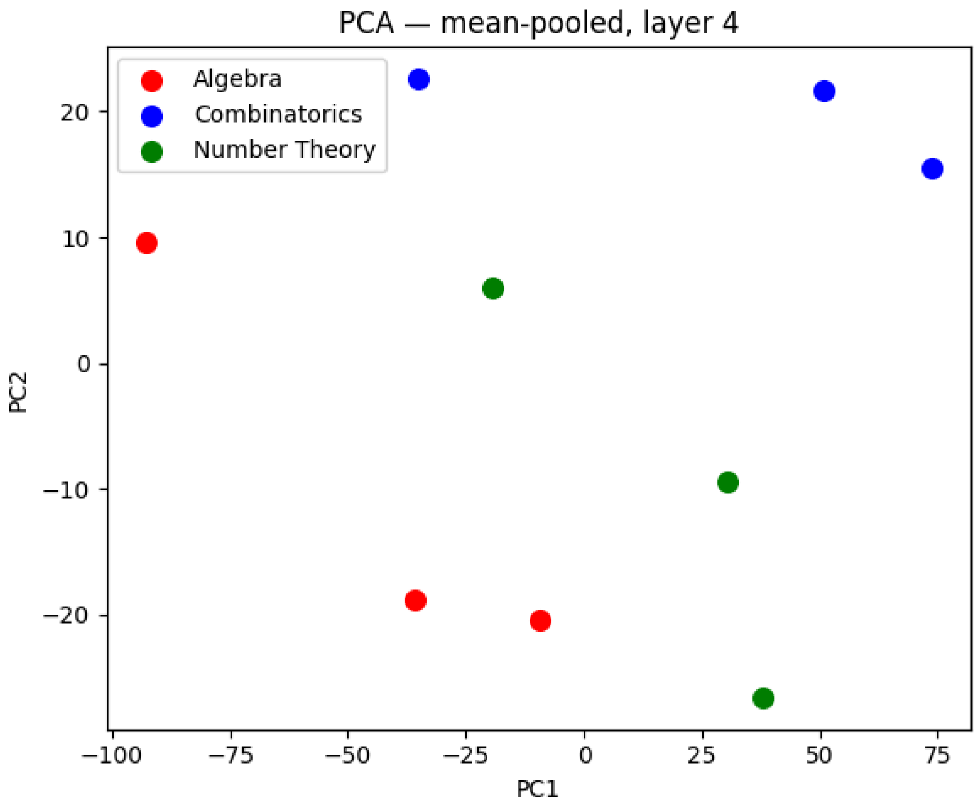
After evaluating performance results, we went with a distilled Deepseek math model `'deepseek-ai/DeepSeek-R1-Distill-Qwen-7B'` as its ability on math is sufficient for our uses, and allows for hidden state capture. In order to investigate the relative signal strength and separability of concepts within hidden states, we ran PCA on all the layers of the model, as well as separate slices. Certain layers were found to have more separability (Layer 4), and we tried several subsets of these activations as supported by the literature [2].

I. Hidden State Initial Analysis

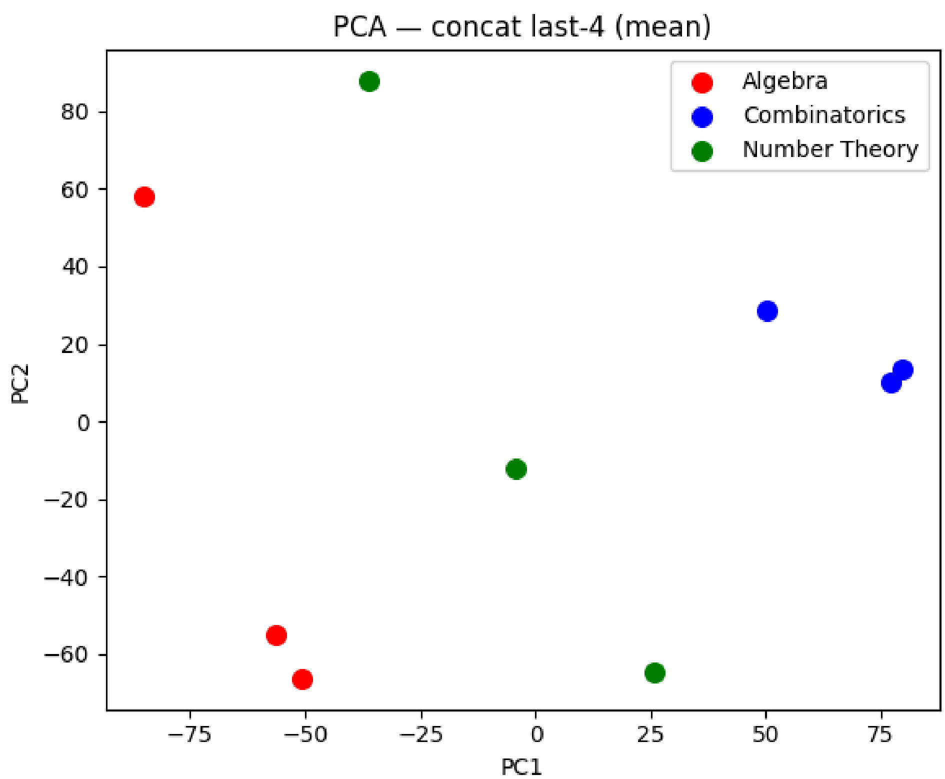
Top layers by LOO-LogReg accuracy:

Layer	Token Transform	Accuracy
4	mean	0.778
10	last	0.778
0	mean	0.667
1	mean	0.667
1	last	0.667
5	mean	0.667
10	mean	0.667
11	mean	0.667

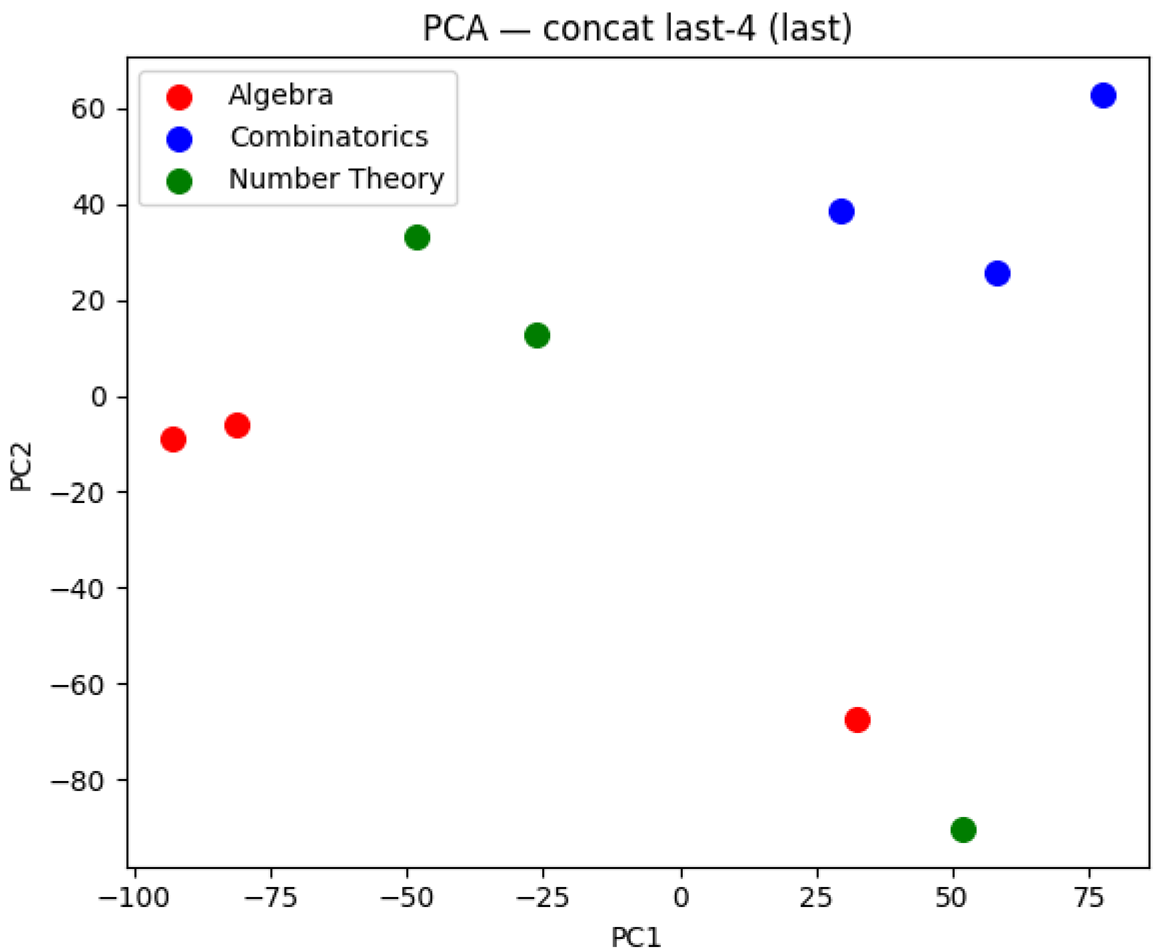
With a very small sample-size, there does exist *some* separability as seen in the PCA plot.



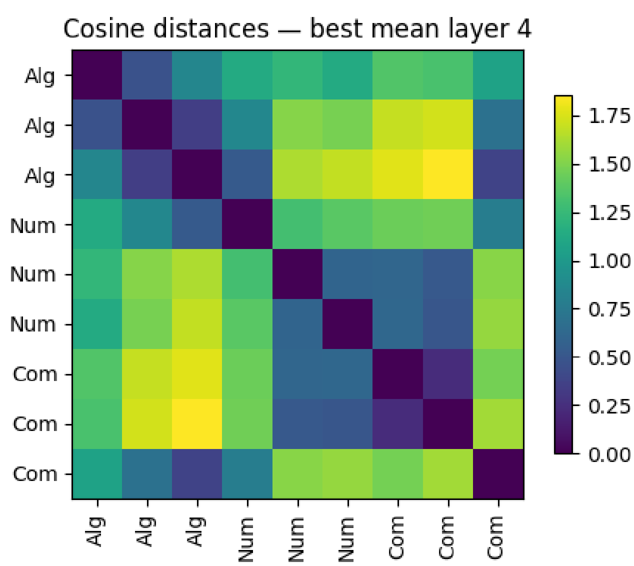
To try and define further separability, (Alain, Bengio 2018) [5] mention concatenating the last 4 layers (as these layers seem to contain denser topic information), and does seem to be the case.

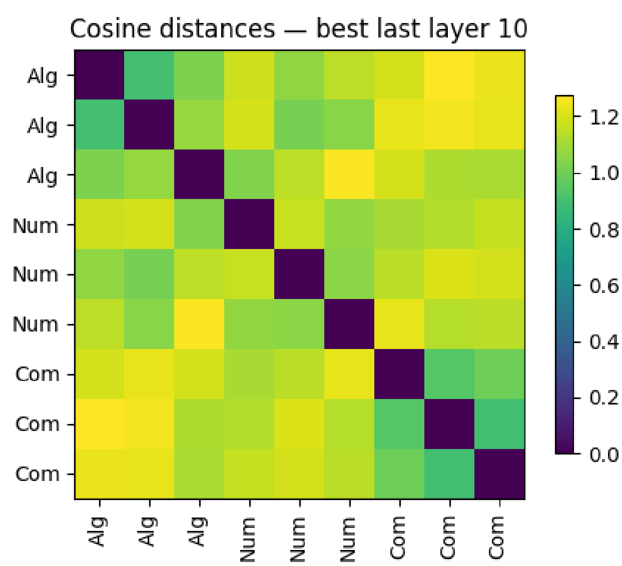


Though the last token activation, which ideally contains dense information, doesn't hold for all cases. As seen here, where Algebra and Number Theory points are relatively close (not rigorously tested though).



Cosine Distances for the topic probe are giving somewhat mixed results:

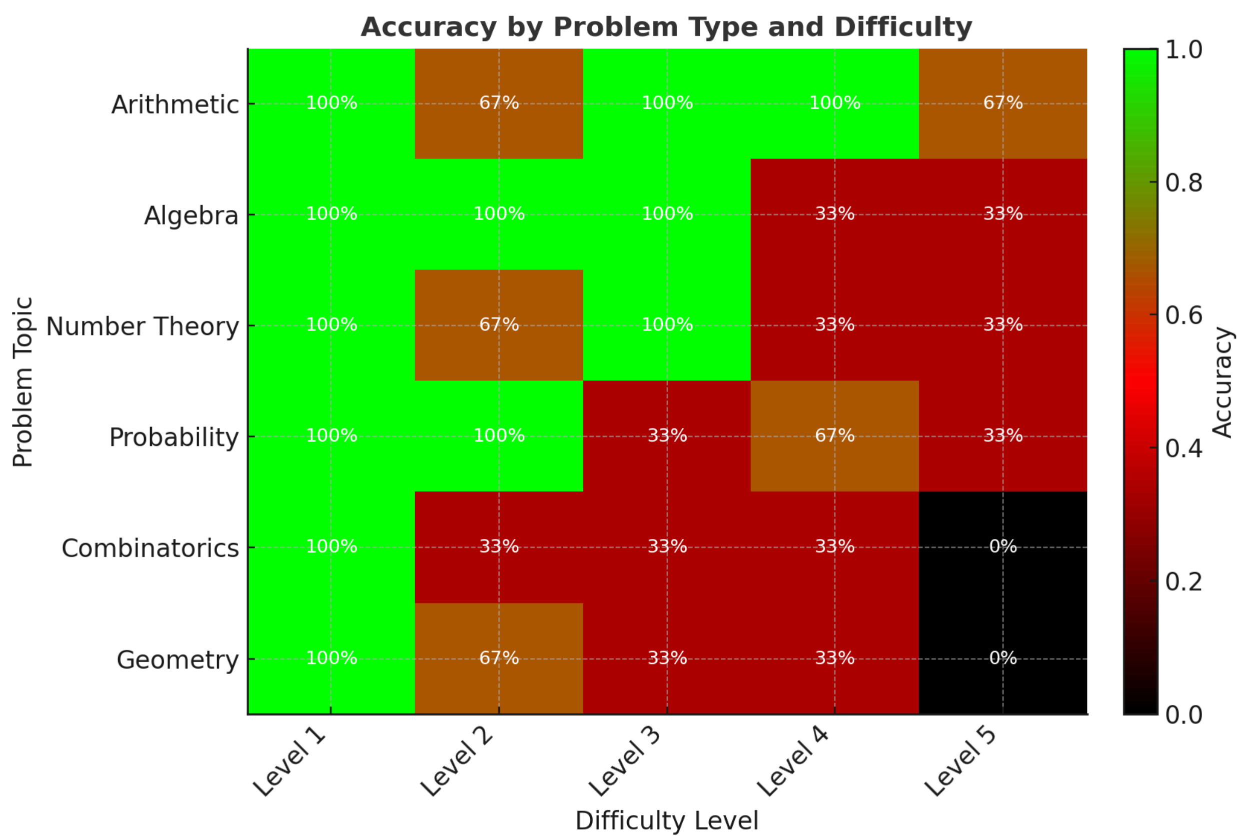




Layer 4 seems to be the best single layer for topic separability, but the average over the concatenated last 4 layers is found to be more separable. Deciding on the linear probe to train on (with limited compute hours) given this means we could train on either a concatenated

II. Initial Performance Testing

We tested a subset of the MATH dataset (75 problems), while capturing hidden states, and tried 3 problems of each difficulty x topic. Due to deepseek's /think behavior when answering questions, we ran into many issues with model repetition, and non-adherence to the {boxed} behavior we ask it to evaluate with. Simple parsing for the {boxed} answer was very problematic, and we will need a more robust way to evaluate correctness of the model's response without blowing up the resources needed. Prompting the model to not use 'think' or early stopping was tried, but manual evaluation of model response found that these strategies can result in false negatives. With our noisier parsing of model responses, we found this performance with the model:



Notably, **Geometry** is the worst performing category, while **Arithmetic** seems to be the best. Geometry shows the sharpest decline with difficulty, emphasizing the model's struggle with spatial and diagram-dependent reasoning. Other subjects degrade more gradually, suggesting increasing algebraic or multi-step reasoning complexity rather than conceptual breakdowns. A trend exists within difficulties, as expected, but this trend is noisy and not something necessarily to index on with such low samples.

The success probe was not able to be implemented before this report due date, but we aim to get the 0% success probe trained, along with the topic probe trained using layer 4, given it was the most separable layer for topics.

6. Next Steps

So far, we have prepared a subset of the MATH dataset by topic and difficulty, and verified model readiness for activation capture. Our immediate focus is to move from data and model setup to the probing and analysis stages outlined in our proposal.

In the upcoming phase, we will:

- **Implement and train the Topic Probe:**
  - Extract hidden states after question encoding.
  - Train linear classifiers to test topic decodability (algebra, number theory, combinatorics).
  - Perform a layer sweep to identify where topic information is most linearly separable.
- **Develop and evaluate the Success Probe:**
  - Generate and label problems with correctness outcomes.
  - Capture activations at 0% (question-only) and 50% (mid-solution) checkpoints.
  - Train logistic regression probes to assess whether model activations encode success likelihood.
- **Implement the Difficulty Probe (if time permits, this may be removed so we can reduce scope and make our goals more achievable)**
  - Predict human difficulty level from question-only representations.
  - Correlate predicted difficulty with empirical model accuracy.
- **Conduct initial analyses and visualizations:**
  - Compare probe performance across layers and objectives.
  - Relate probing signals to accuracy trends across topic × difficulty.
  - Prepare early figures and insights for inclusion in the final report.

Near-term goals:

1. Wrap up model setup and test it using a small subset of the MATH data
2. Complete implementation and evaluation of the topic and success probes.
3. Generate layer-wise results and interpret emerging trends.
4. Begin drafting visual and written components for the final paper.

References

[1] **Marks, S., Tegmark, M.** *The Geometry of Truth: Emergent Linear Structure in Large Language Model Representations of True/False Datasets.* arXiv:2310.06824, 2023. (arXiv)

[2] **Vicente Moreno Cencerrado, I., Padrés Masdemont, A., Gonzalvez Hawthorne, A., Africa, D. D., Pacchiardi, L.** *No Answer Needed: Predicting LLM Answer Accuracy from Question-Only Linear Probes.* arXiv:2509.10625, 2025. (arXiv)

[3] **Hendrycks, D.,** et al. *Measuring Mathematical Problem Solving With the MATH Dataset.* arXiv:2103.03874; NeurIPS Datasets and Benchmarks 2021. (arXiv)

[4] **Wang, X.,** et al. *Self-Consistency Improves Chain of Thought Reasoning in Language Models.* arXiv:2203.11171, 2022. (arXiv)

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